ETH zürich

Information Retrieval #10 - Evaluation

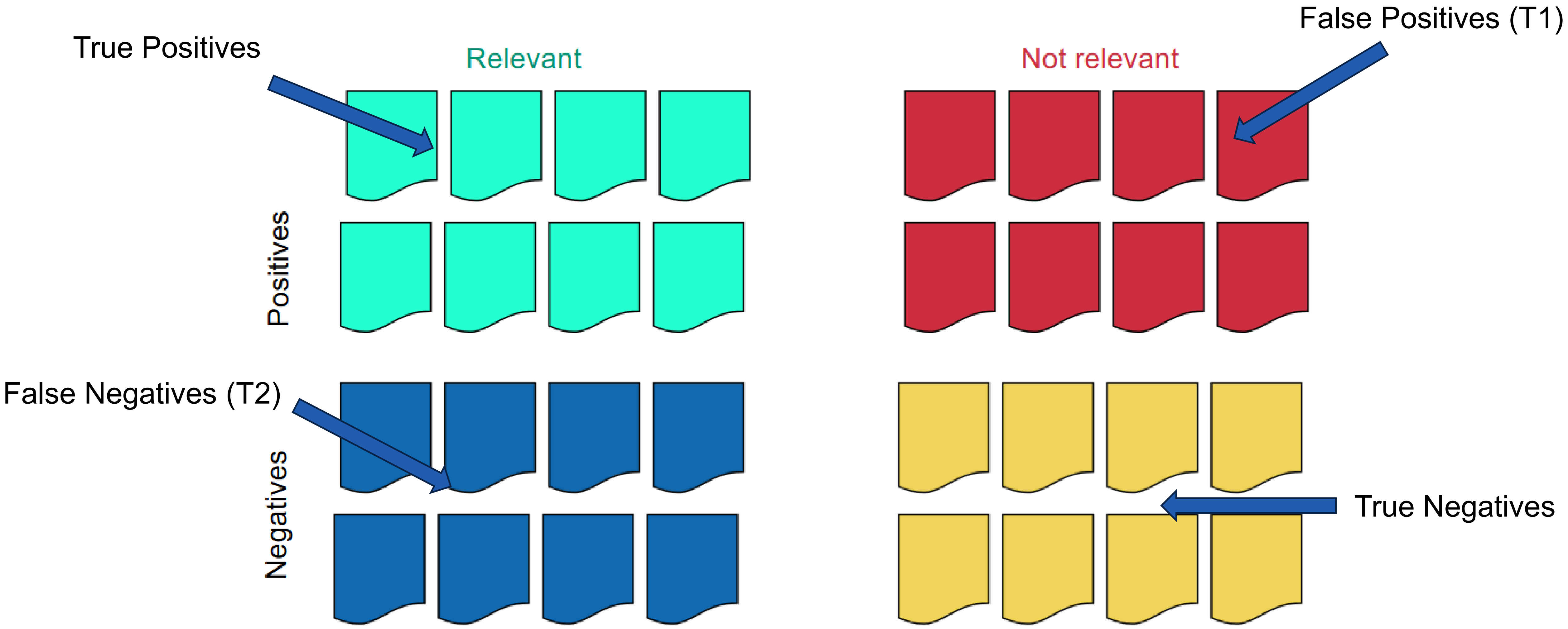
Severin Mills

15.05.2026

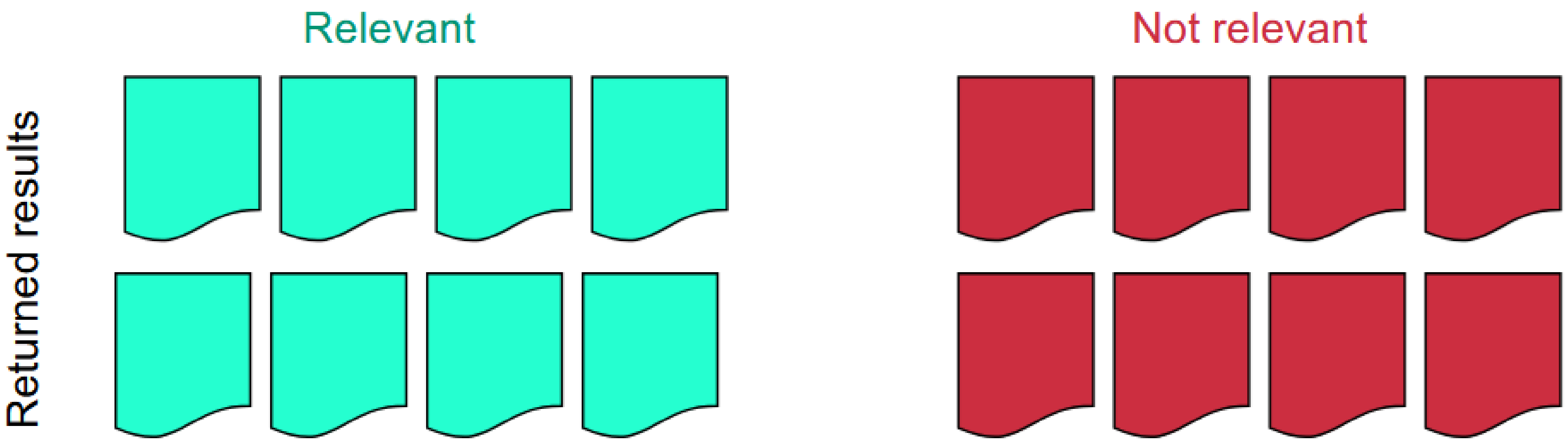
Agenda

1. Lecture Recap
2. Exercise 9: Evaluation
3. Exam Questions
4. Kahoot

Evaluation, Previously…

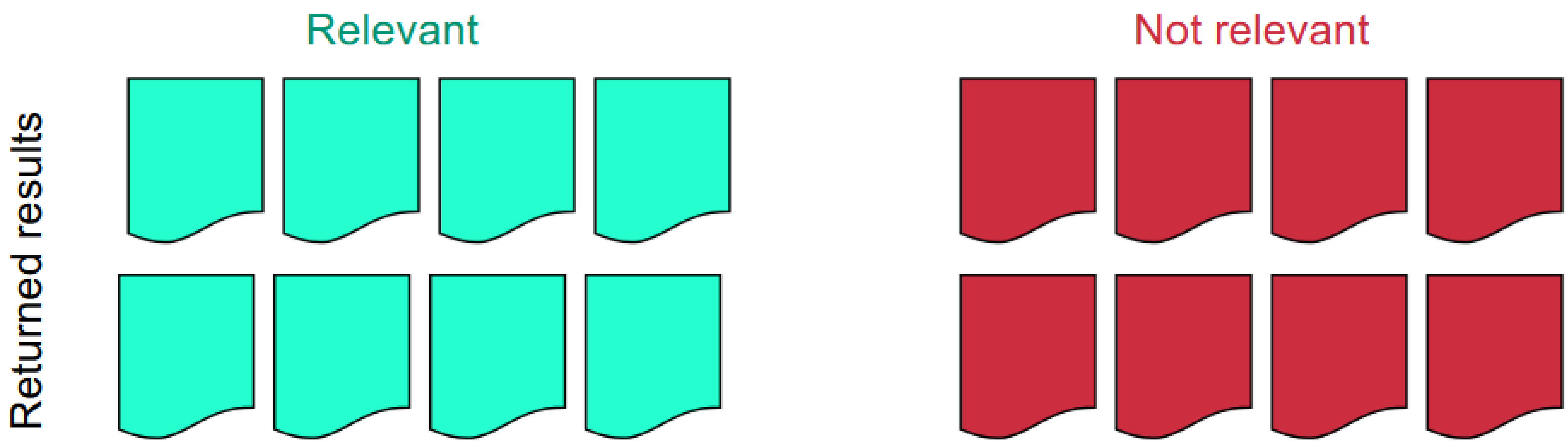


Precision



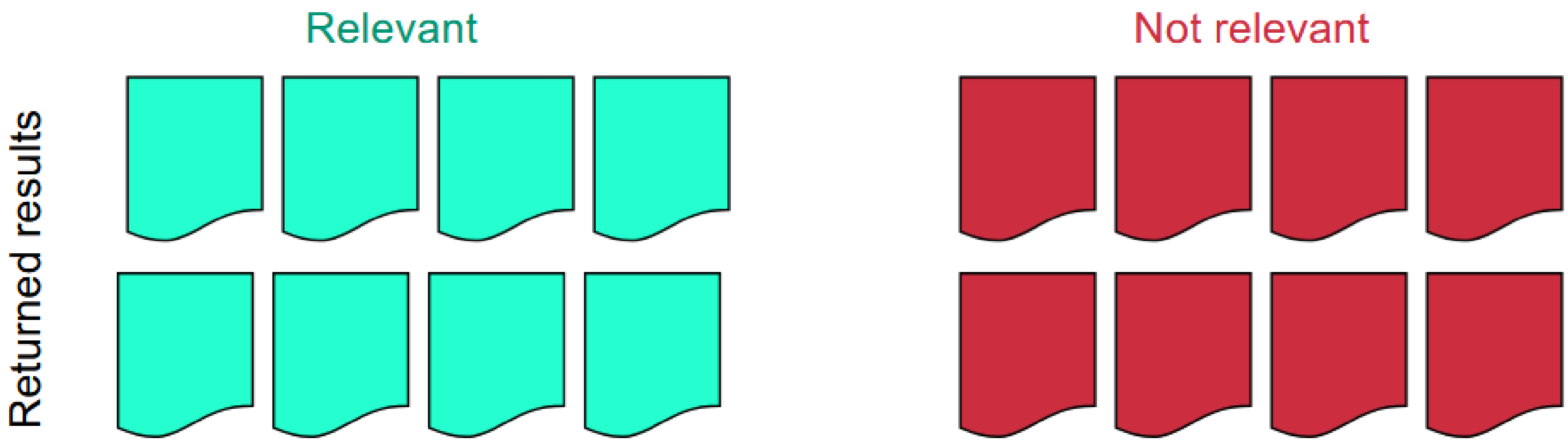
$$\text{Precision} = \frac{\text{true positives}}{\text{returned}}$$

Precision



Precision =

Precision



Precision = 50%

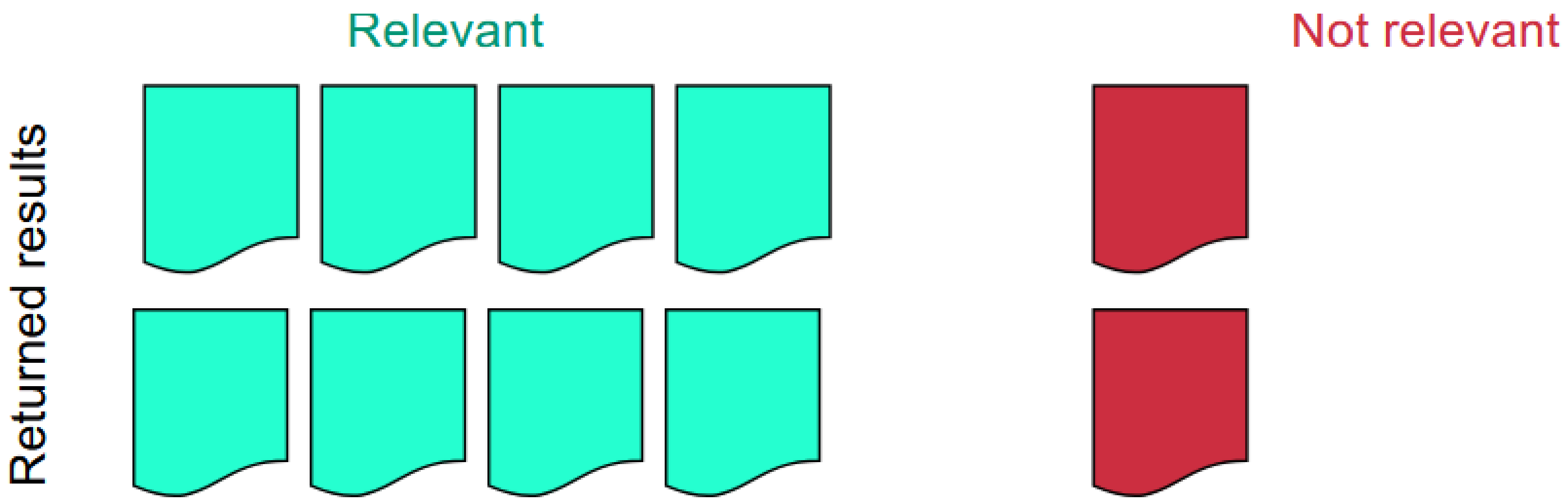
Precision

Precision =

Precision

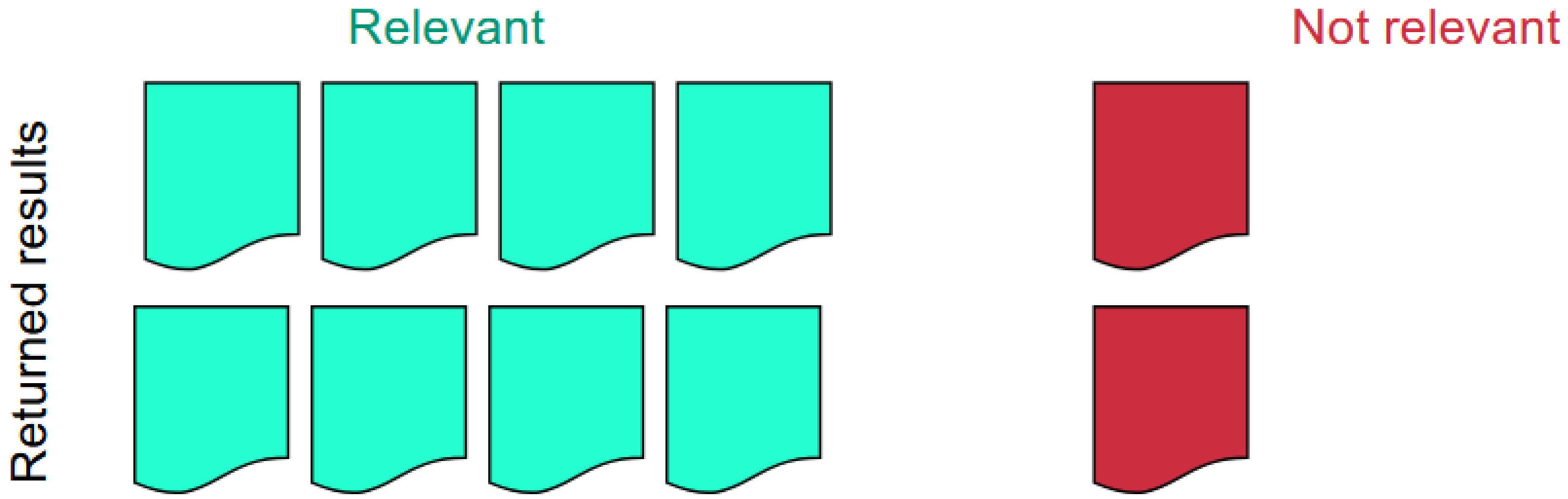
Precision = 0%

Precision



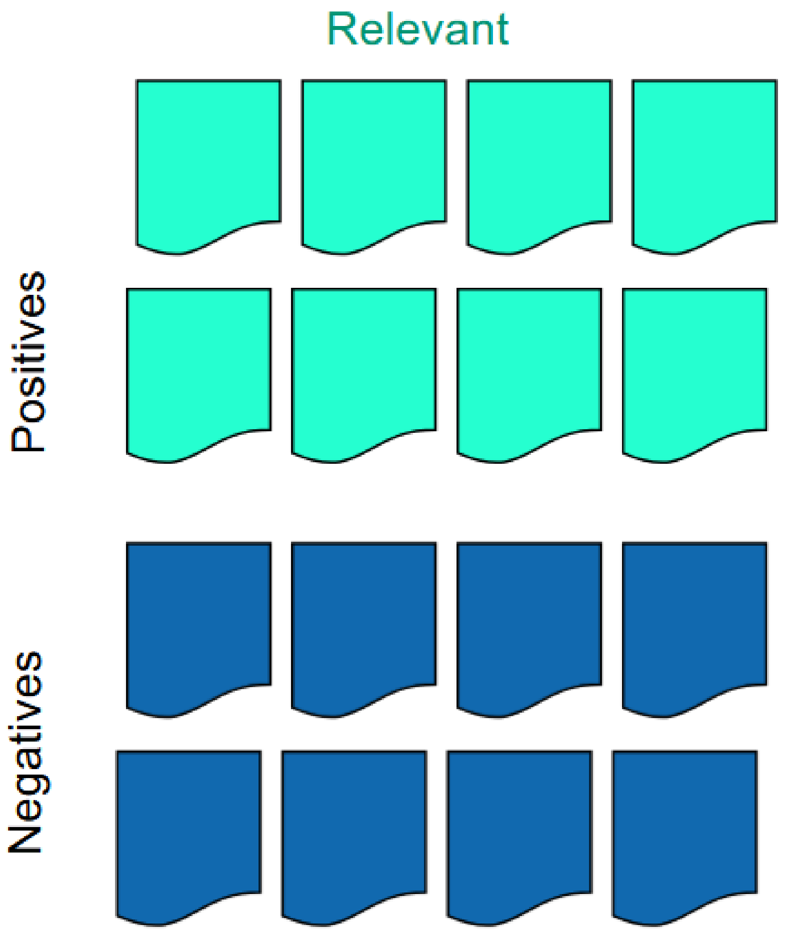
Precision =

Precision



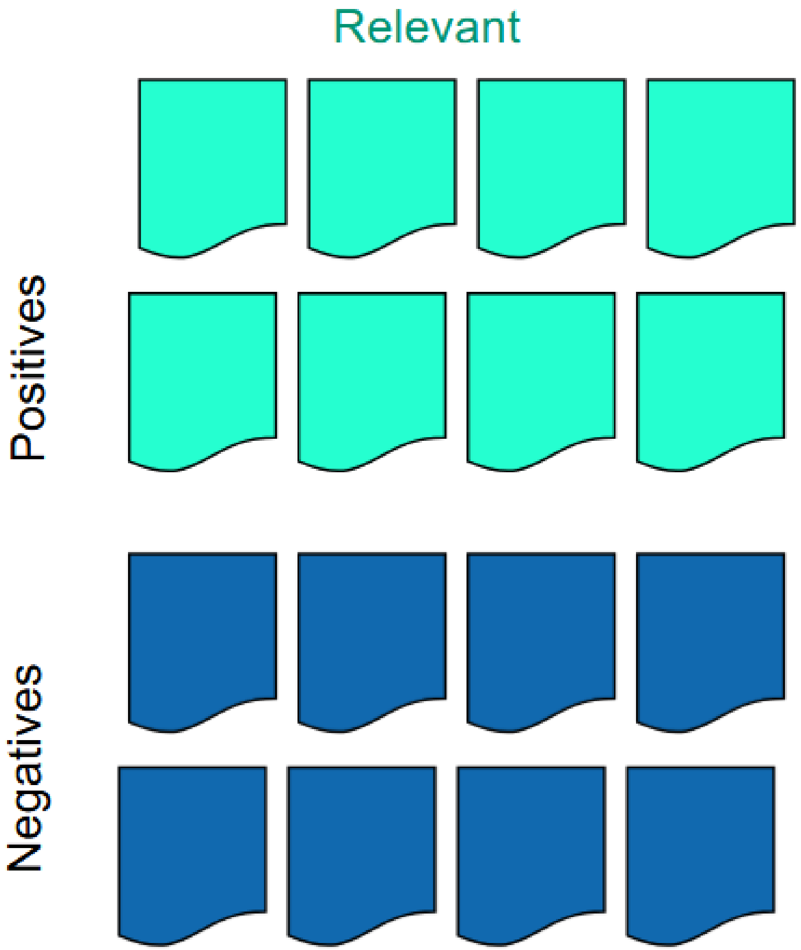
Precision = 80%

Recall



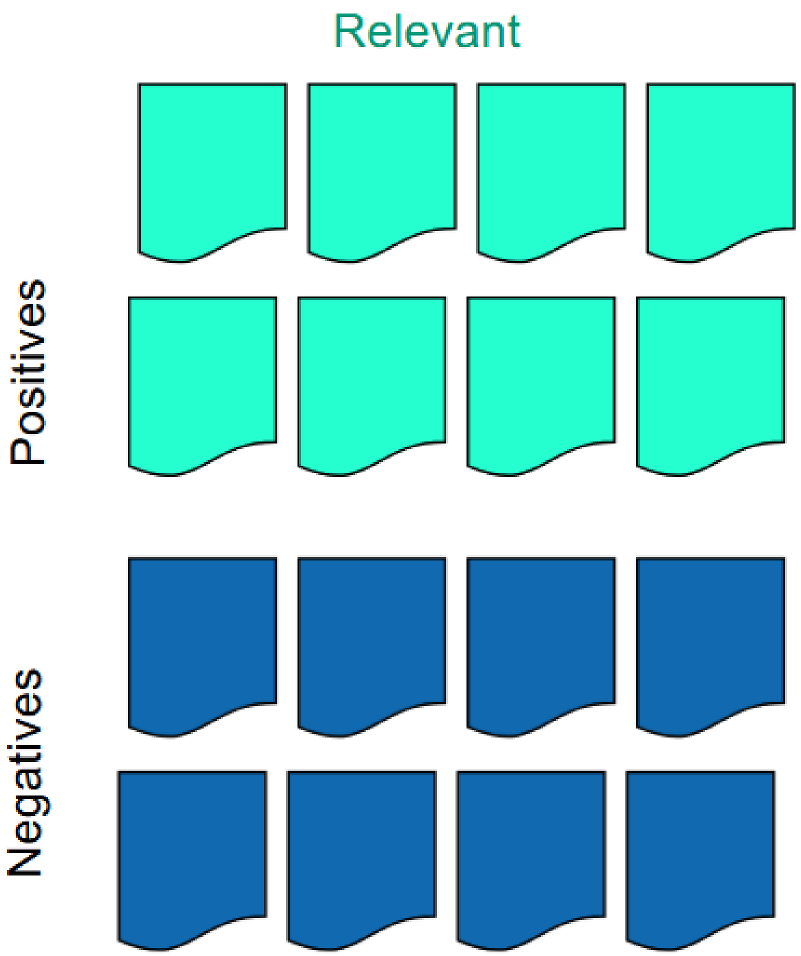
$$\text{Recall} = \frac{\text{true positives}}{\text{relevant}}$$

Recall



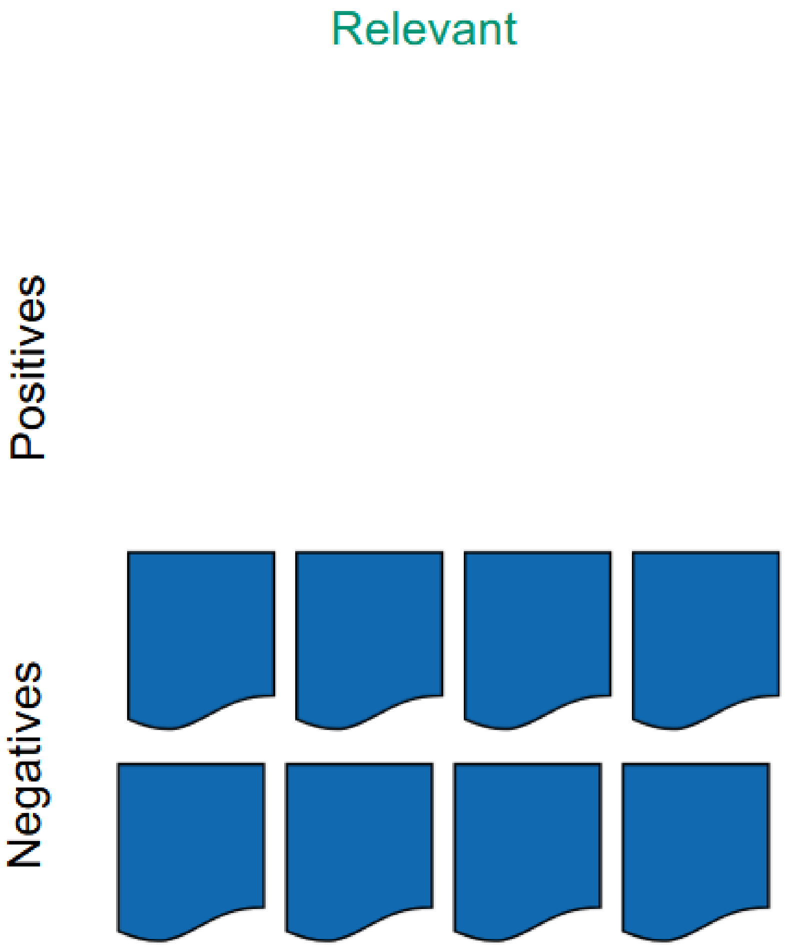
Recall =

Recall



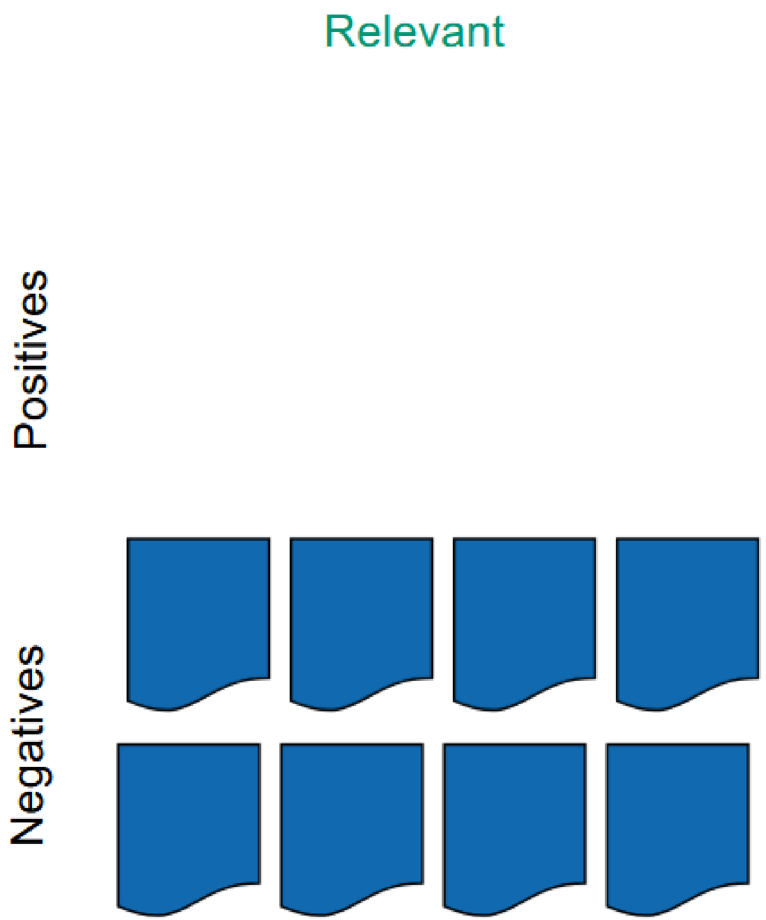
Recall = 50%

Recall



Recall =

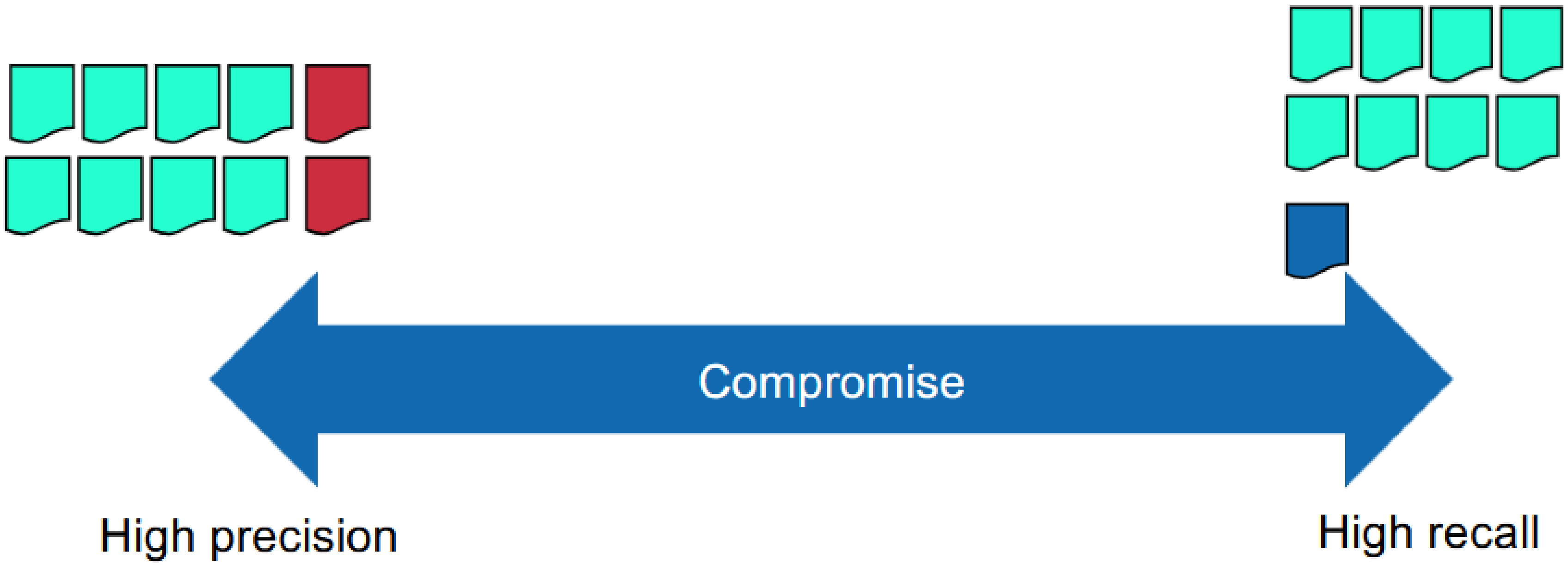
Recall



Recall = 0%

Precision vs. Recall

- Generally, in an IR system, we must compromise between precision and recall



Solution: F-Measure

- Try to “merge” precision and recall to achieve a better evaluation metric

$$F_\alpha = \frac{1}{\frac{\alpha}{P} + \frac{1-\alpha}{R}}$$

$\alpha = 0$

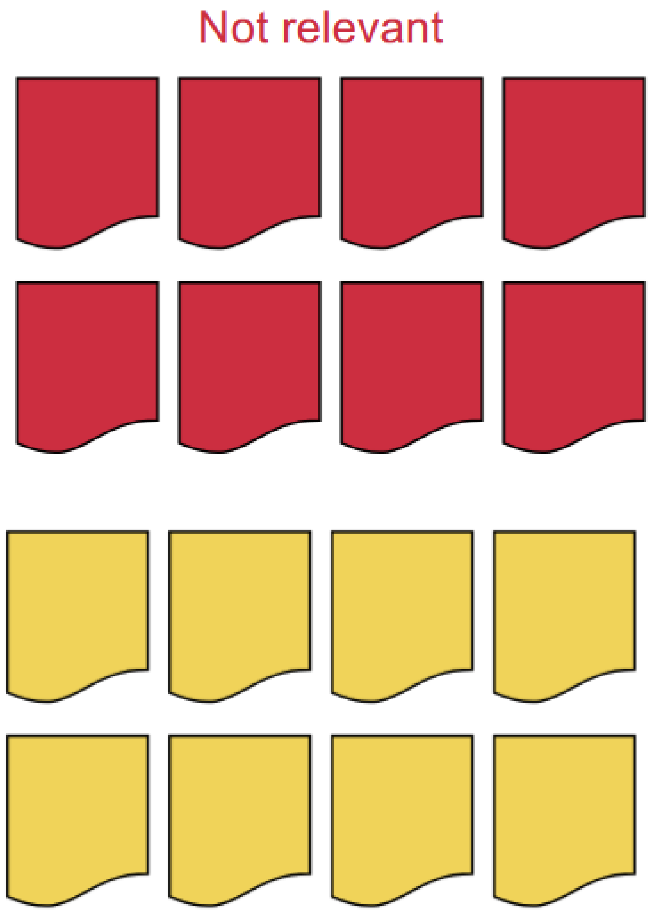
$\alpha = 1$

Recall-heavy

Precision-heavy

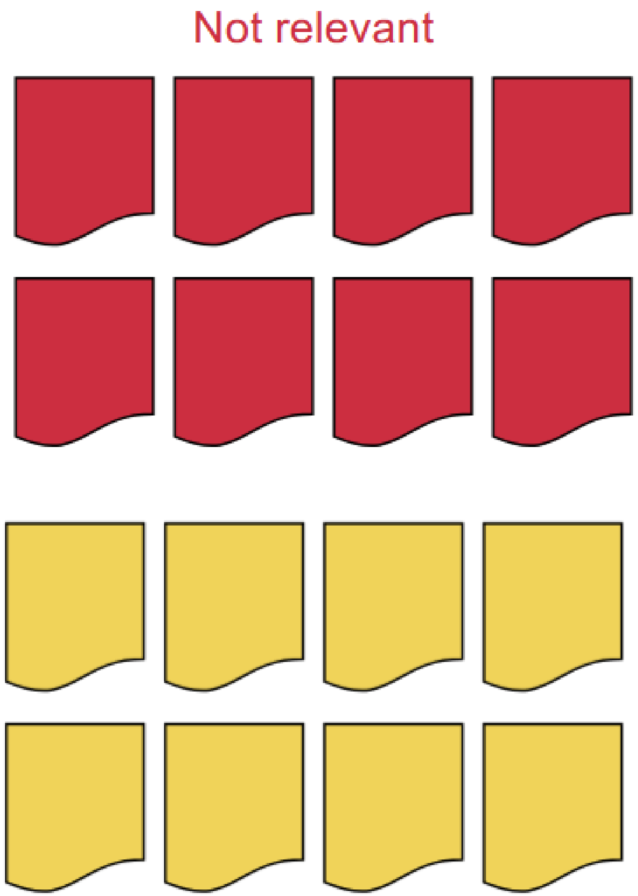
Specificity

$$\text{Specificity} = \frac{\text{true negatives}}{\text{all negatives}}$$



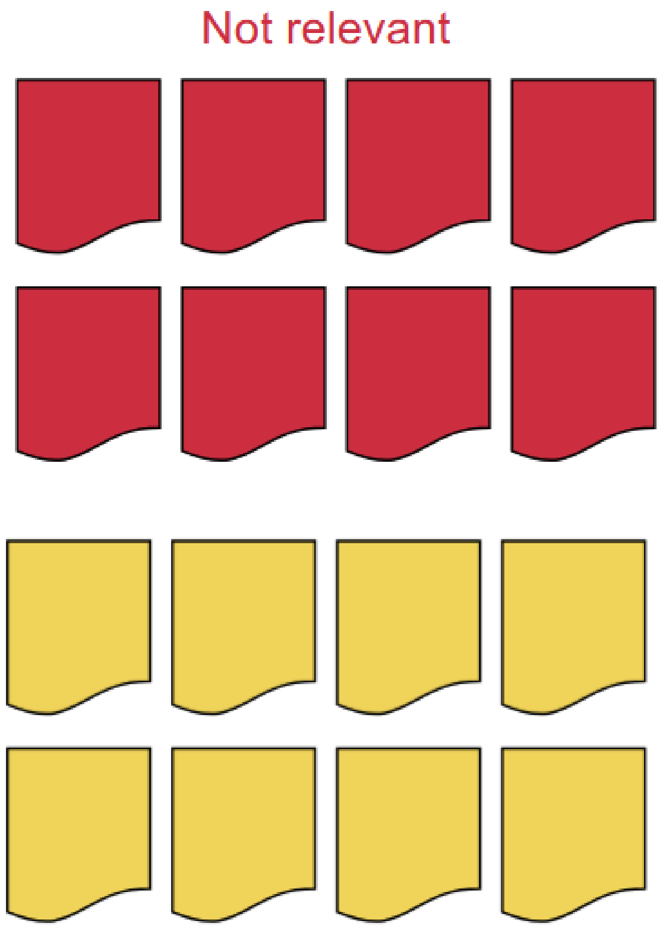
Specificity

Specificity =



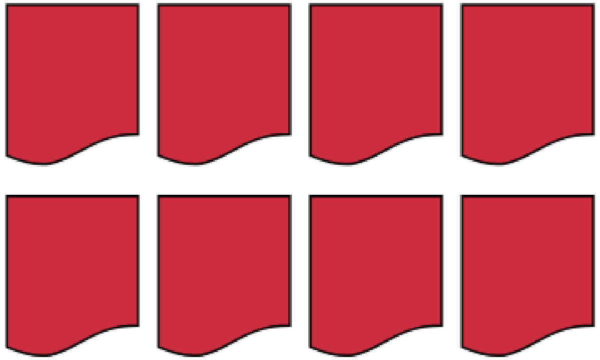
Specificity

Specificity = 50%



Specificity

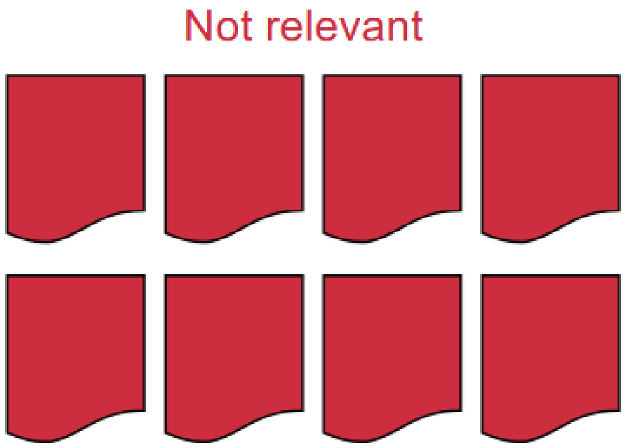
Not relevant



Specificity =

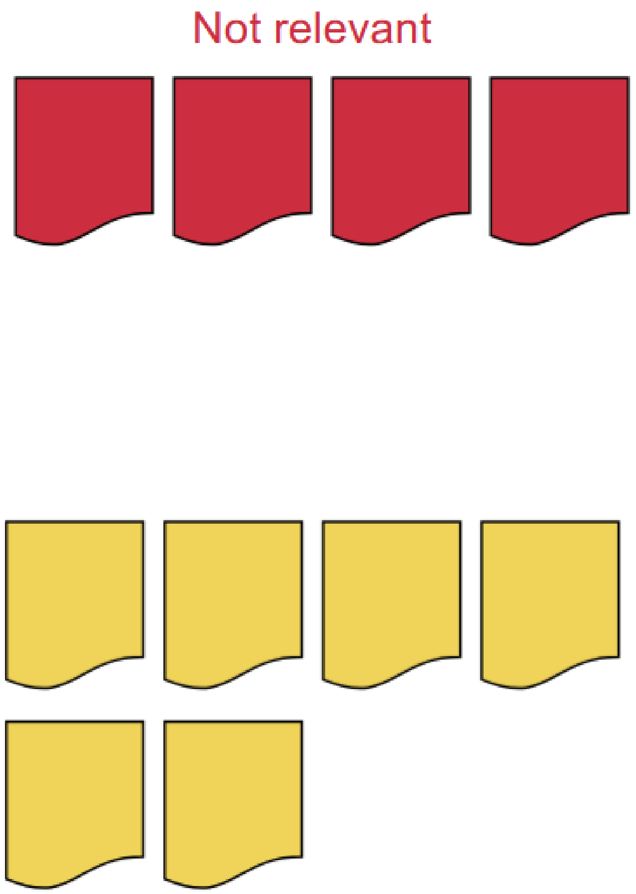
Specificity

Specificity = 0%

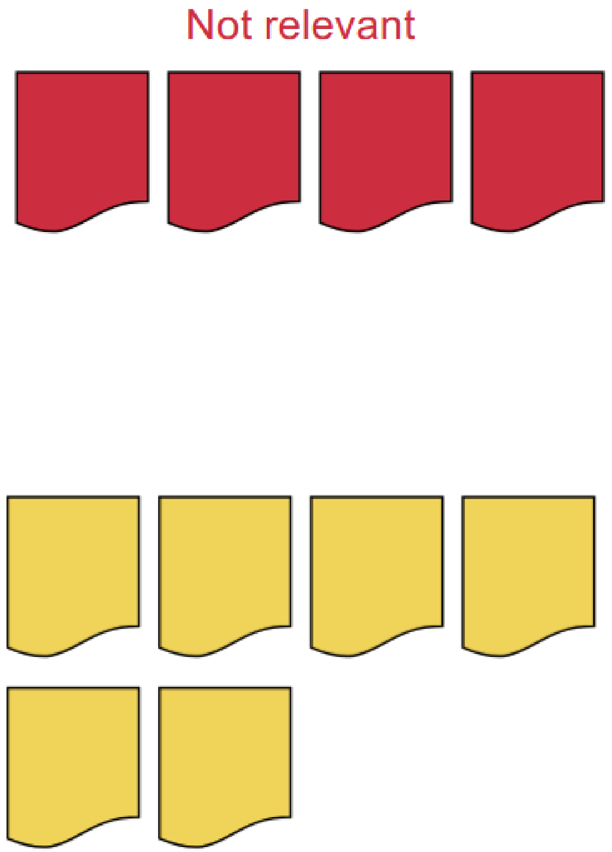


Specificity

Specificity =

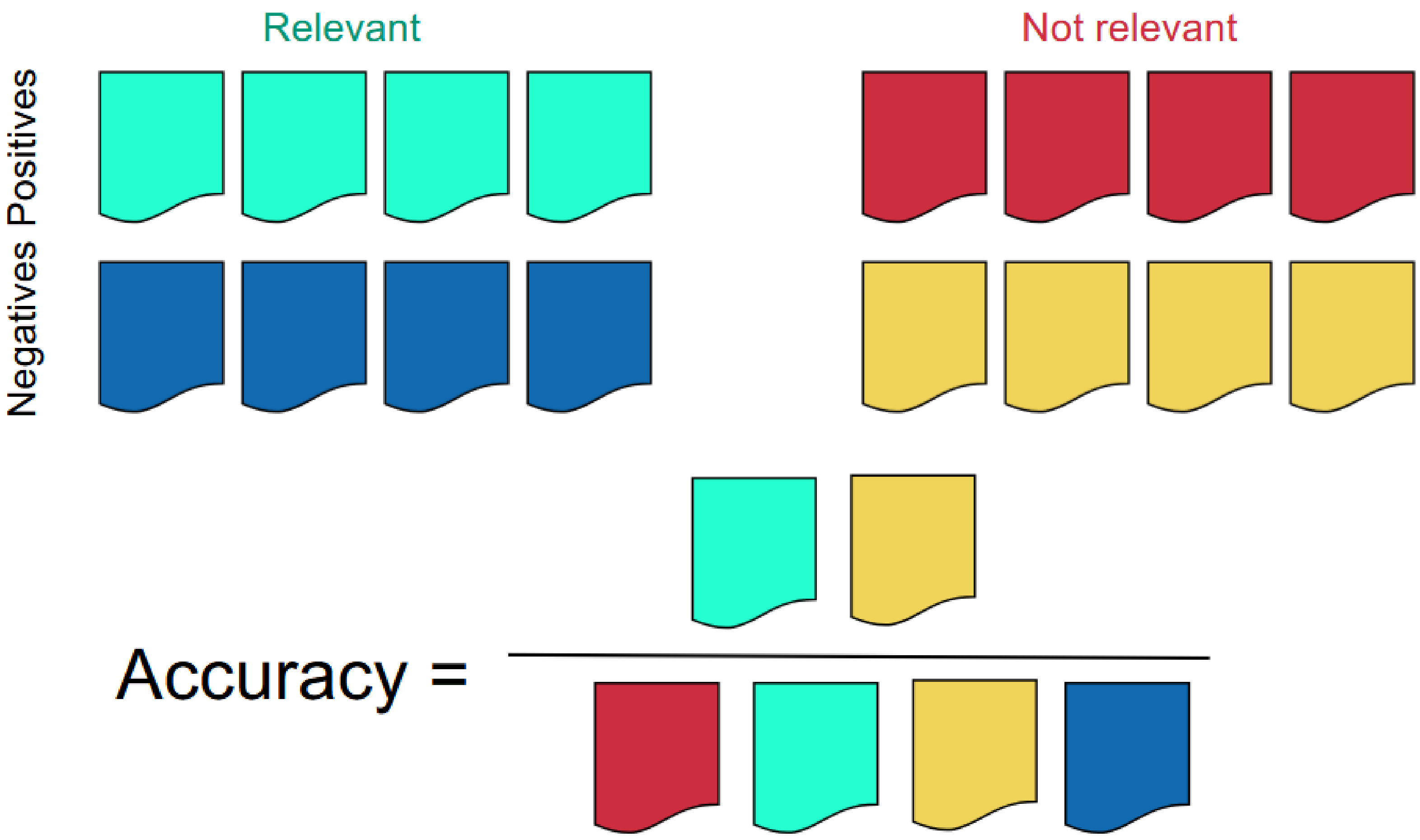


Specificity

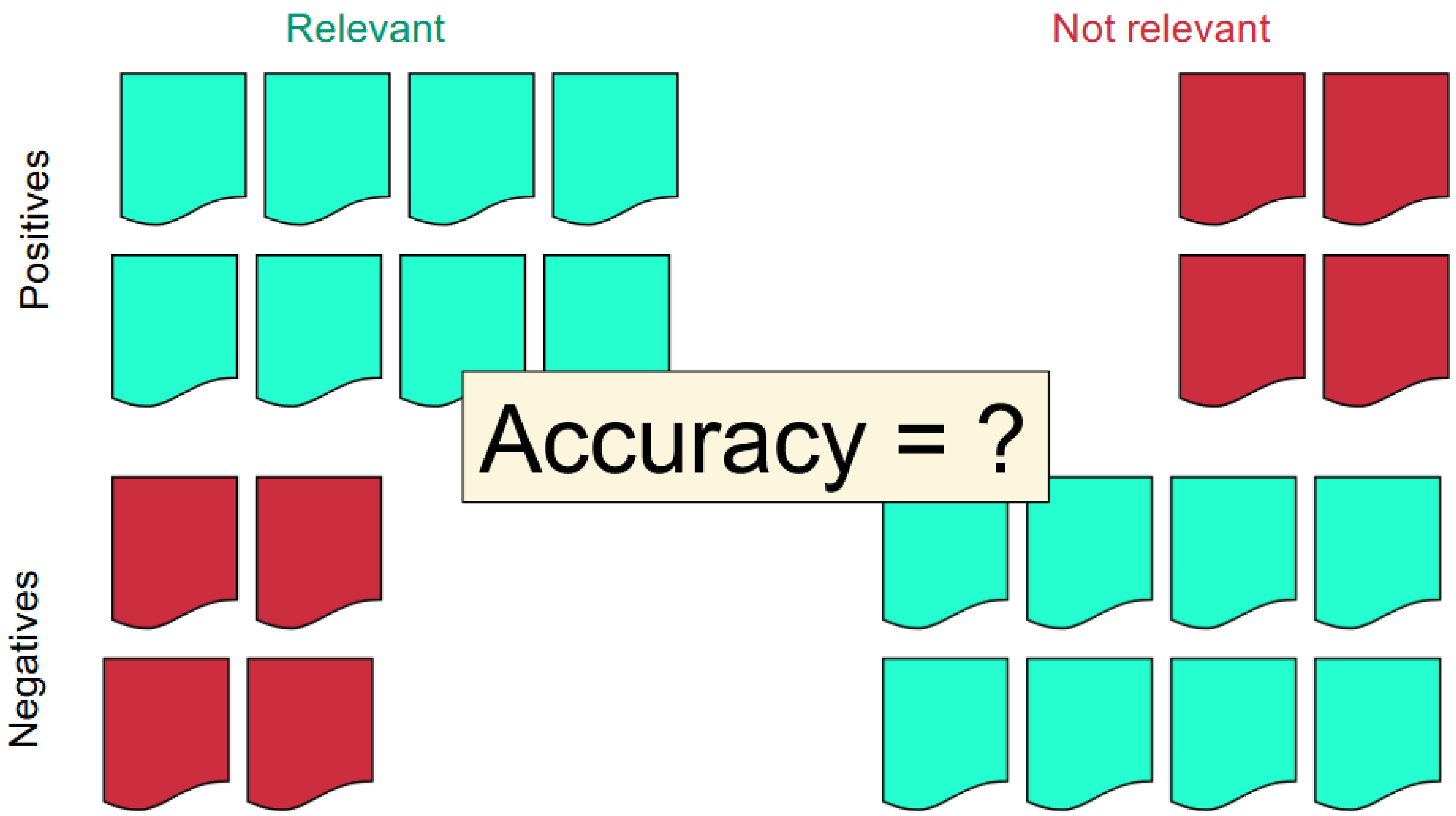


Specificity = 60%

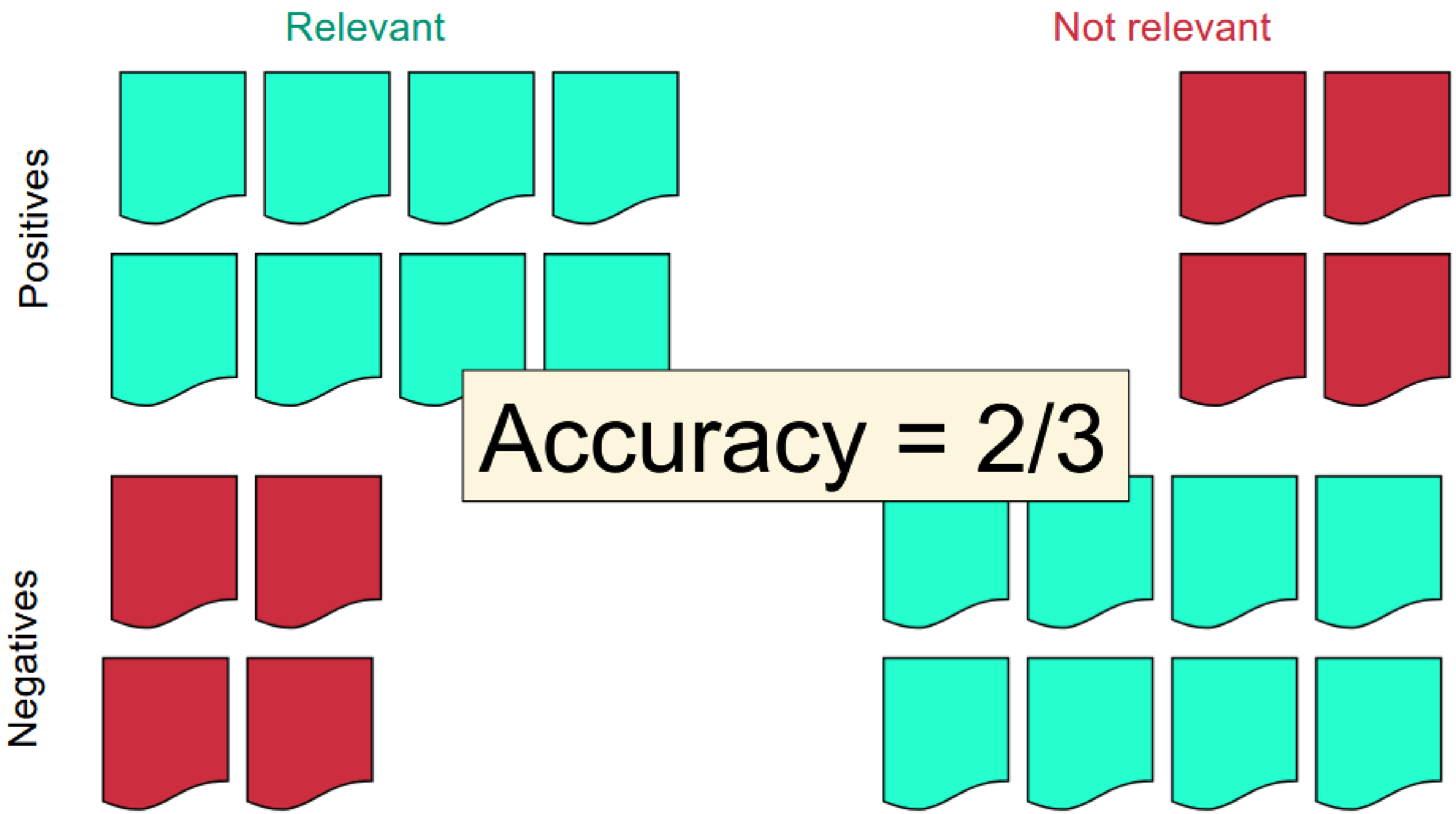
Accuracy



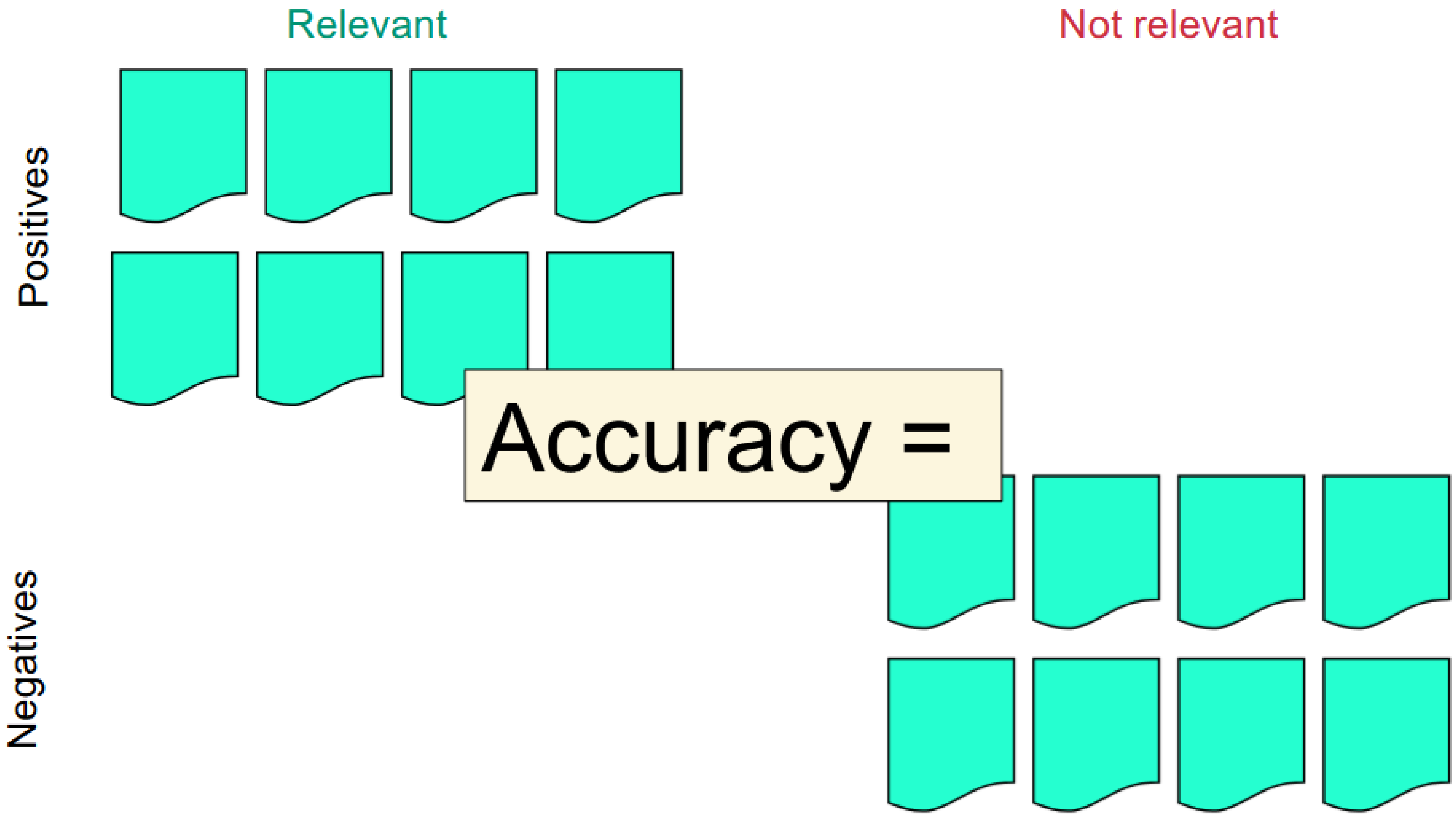
Accuracy



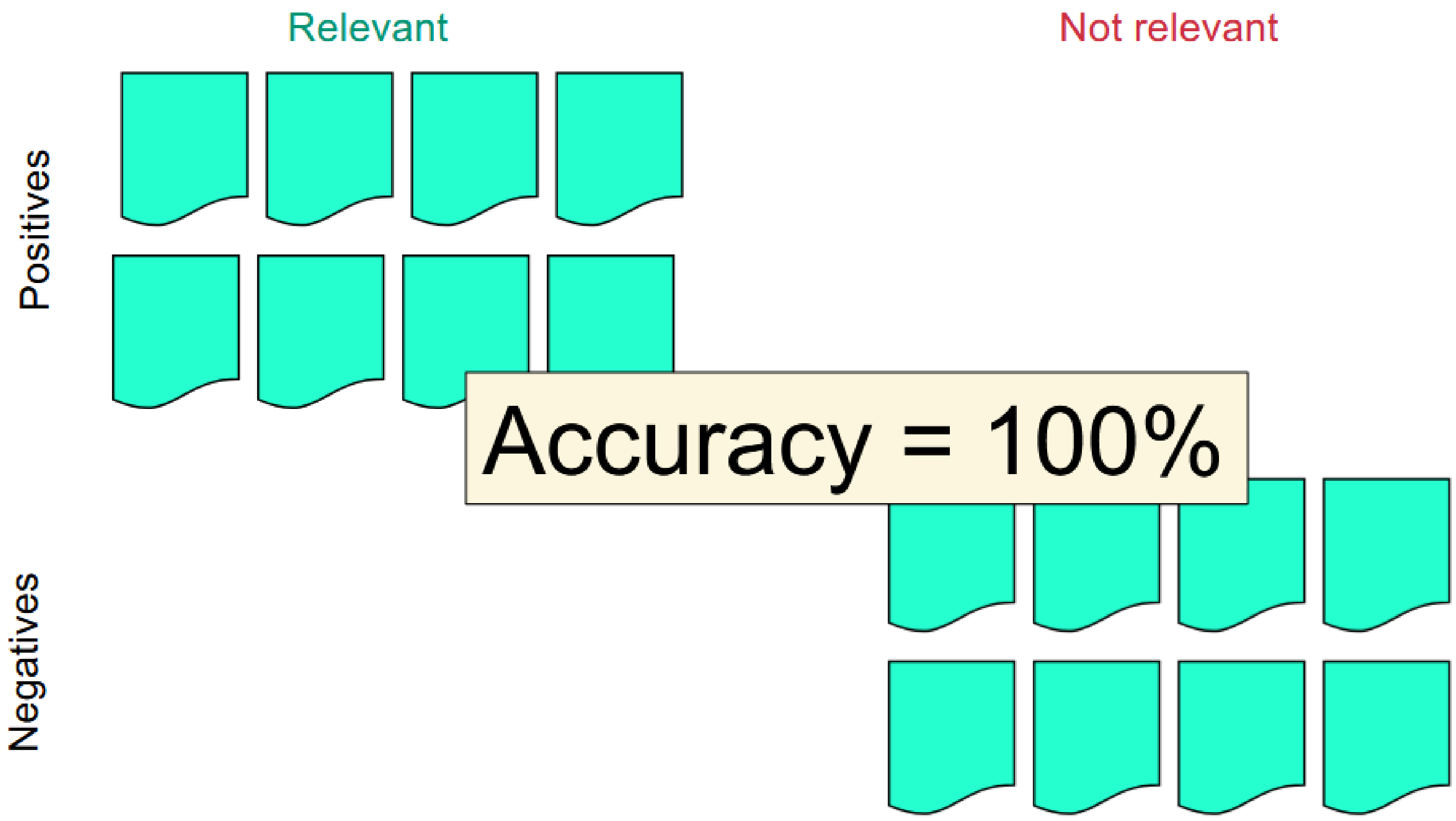
Accuracy



Accuracy

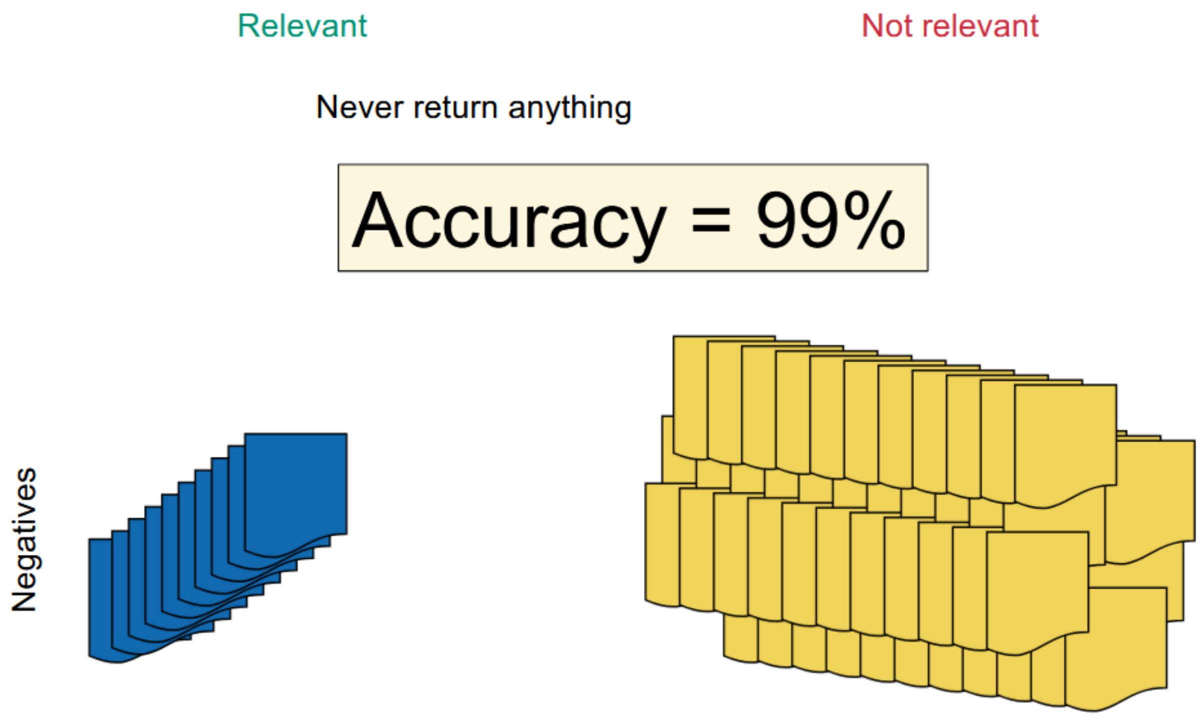


Accuracy



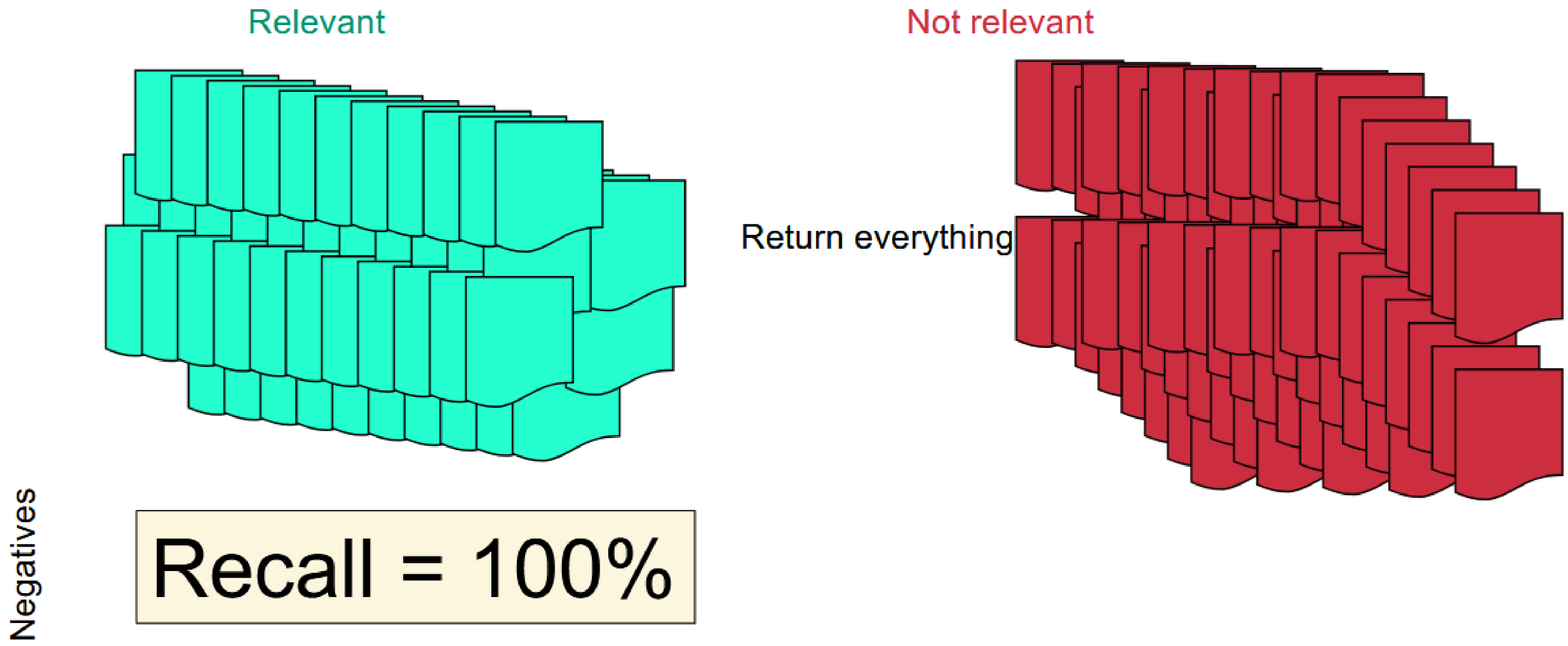
Hacking Accuracy

- If we never return anything, accuracy will be high. Why?
  - Since generally, the set of relevant documents will be much smaller than the set of irrelevant documents



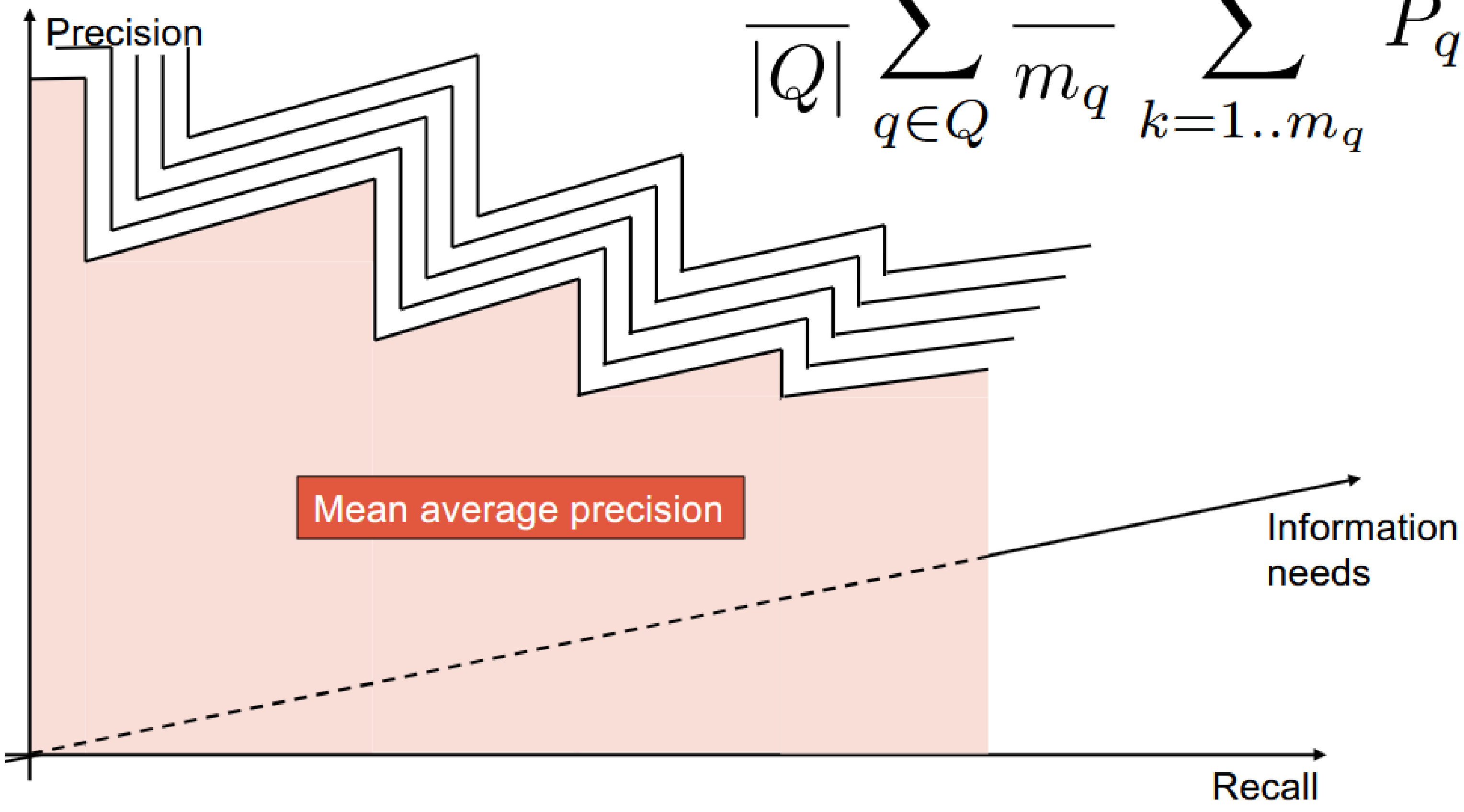
Hacking Recall

- We can achieve perfect recall by returning everything



Mean Average Precision

$$\frac{1}{|Q|}\sum_{q\in Q}\frac{1}{m_q}\sum_{k=1..m_q}P_{qk}\mathrm{rel}_{qk}$$



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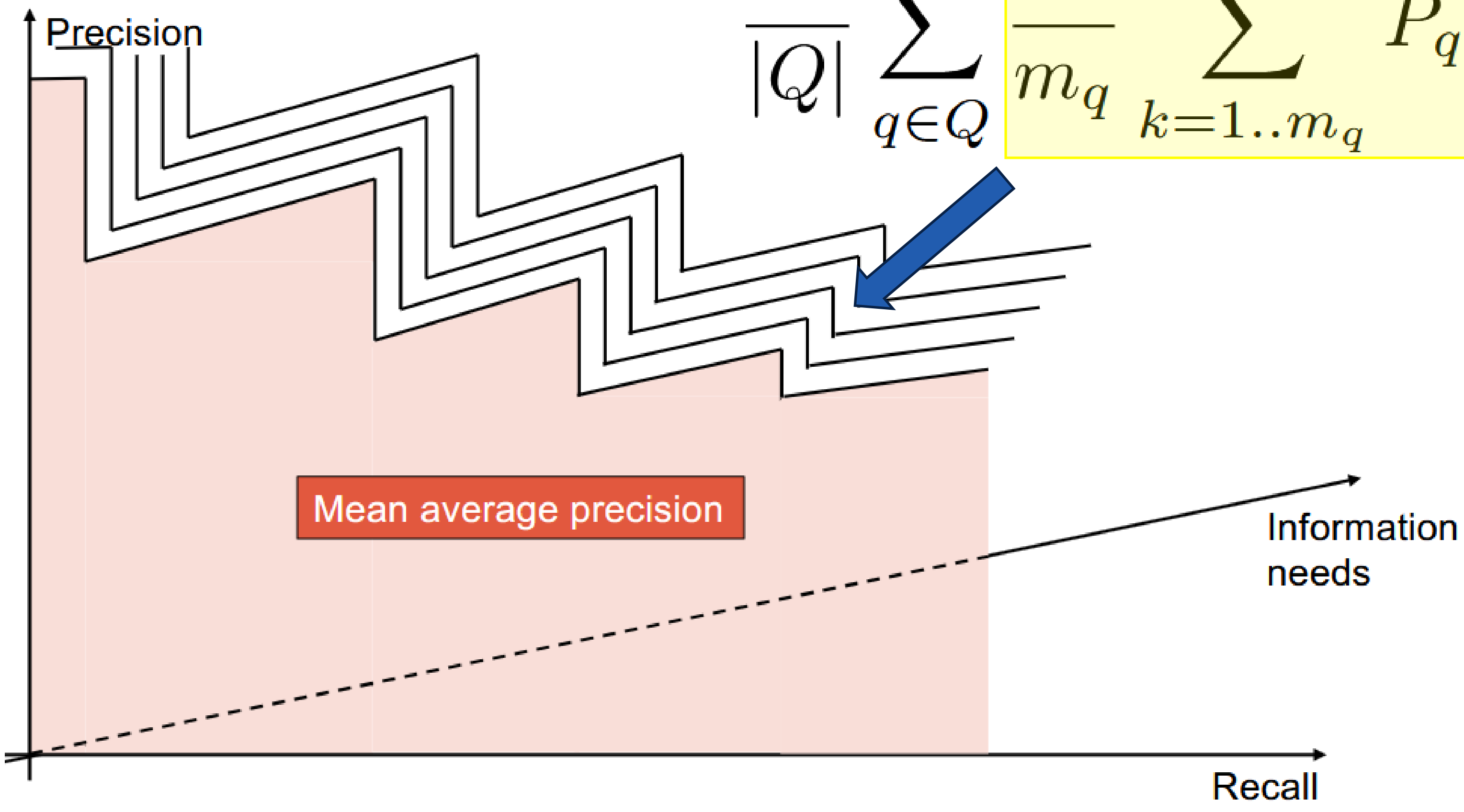

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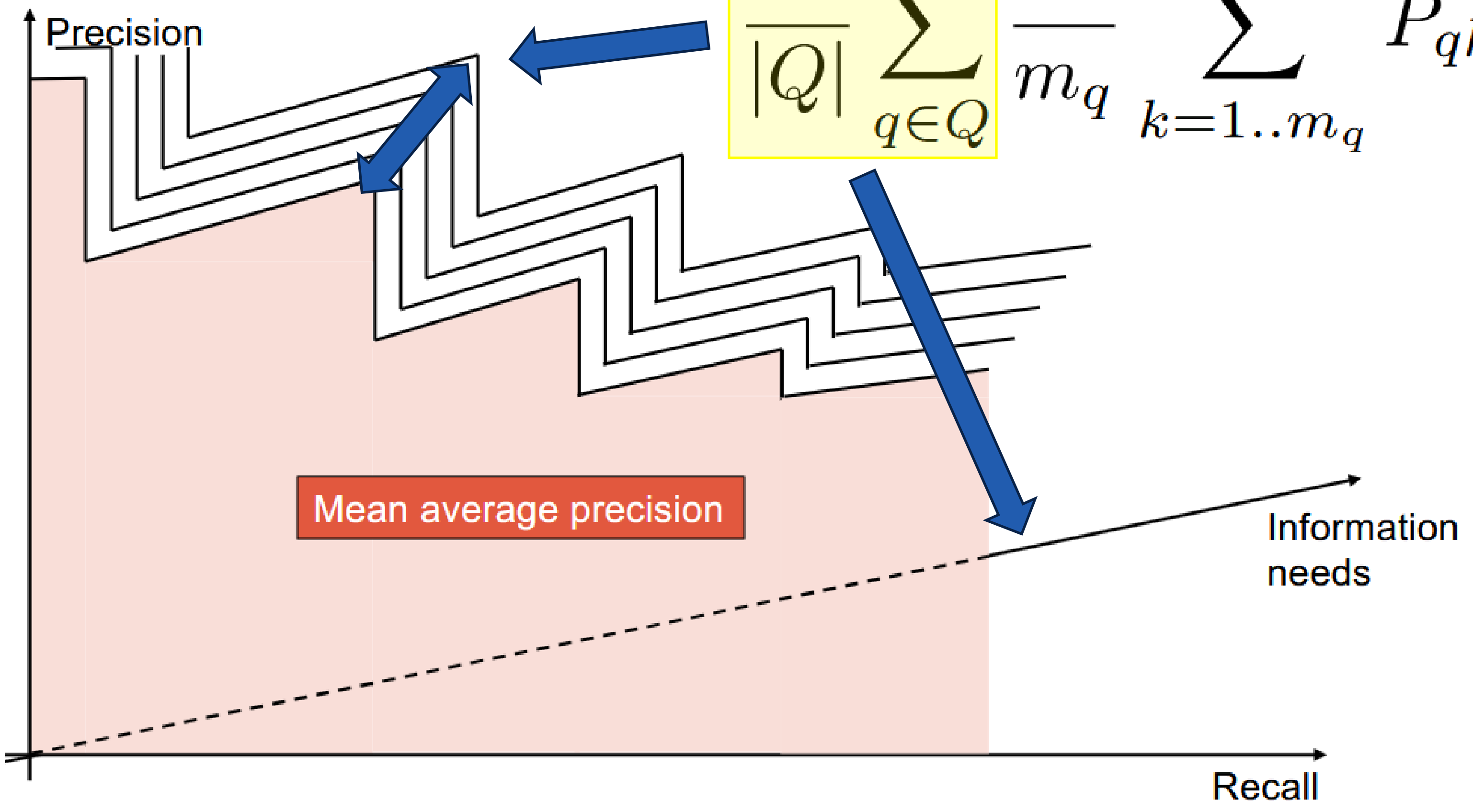
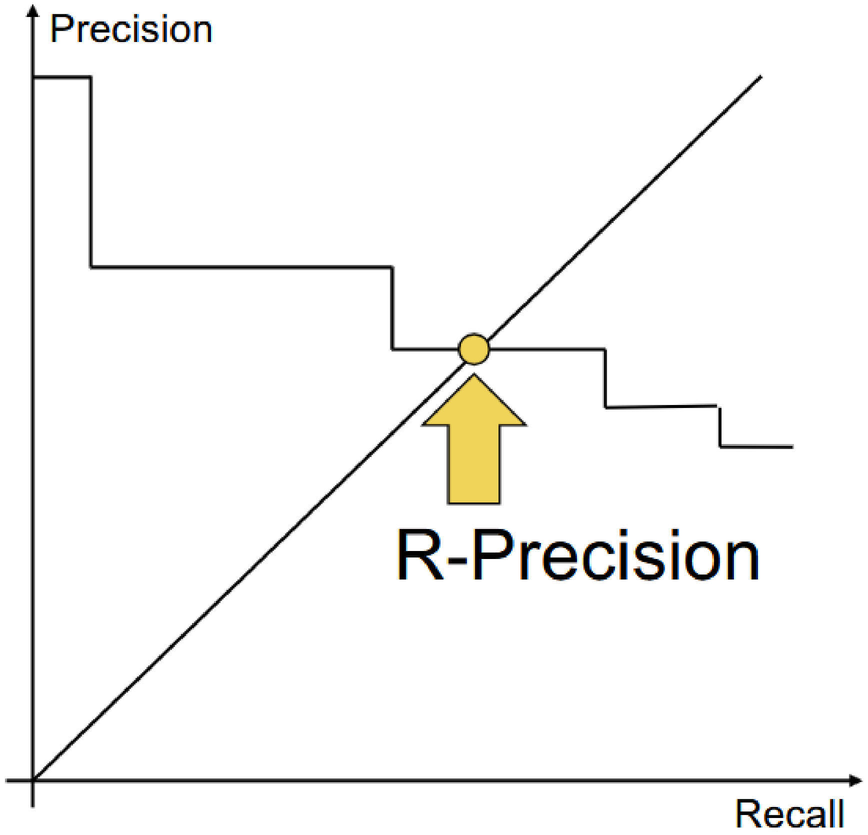
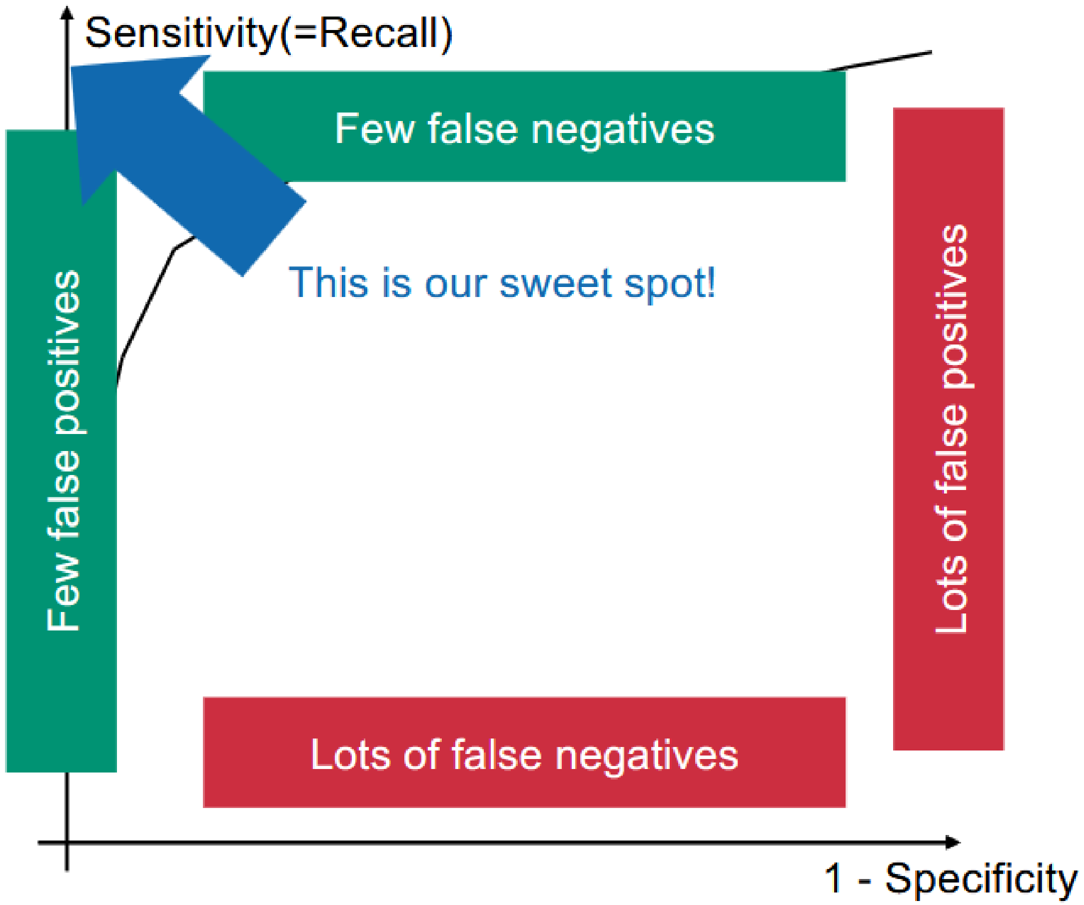


R-Precision

- Idea: We know there are k relevant documents. So, return only the top k results. Then, Precision = Recall!



ROC (Receiver Operating Characteristic) Curve



Exam Questions

- ❑ If the search engine returns nothing, then the accuracy is necessarily 1
- ❑ If the search engine returns nothing, then the recall is necessarily 1
- ❑ If the search engine returns nothing, then the specificity is necessarily 1
- ❑ Although difficult in practice, it is theoretically possible for both the recall and the precision to be 1
- ❑ If the search engine returns everything, then the recall is necessarily 1
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✓ Although difficult in practice, it is theoretically possible for both the recall and the precision to be 1
- True, if we have no FN and FP and only TP, then we get this

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- True, if we have no FN and FP and only TP, then we get this

✓ If the search engine returns everything, then the recall is necessarily 1
- True, we will have no FN, which means recall is 1

❑ If the search engine returns everything, then the specificity is necessarily 1
- False, it is necessarily 0

❑ If the search engine returns everything, then the precision is necessarily 1
- False, there is no distinction between TP and FP, so precision could be anything

Exam Questions

- ❑ The recall is necessarily lower than, or equal to, the precision
- ❑ The precision is always higher than the accuracy
- ❑ If all documents are relevant, then the precision is necessarily 1

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Exam Questions

- ❑ The recall is necessarily lower than, or equal to, the precision
  - False
- ❑ The precision is always higher than the accuracy
  - False. Keeping a precision equal to e.g., ½, it is possible to make the accuracy arbitrarily close to 1 by adding a very large number of TN documents
- ❑ If all documents are relevant, then the precision is necessarily 1

Exam Questions

❑ The recall is necessarily lower than, or equal to, the precision

- False

❑ The precision is always higher than the accuracy

- False. Keeping a precision equal to e.g., ½, it is possible to make the accuracy arbitrarily close to 1 by adding a very large number of TN documents

✓ If all documents are relevant, then the precision is necessarily 1

- True, this means there are no FP, so precision must be 1

Kahoot

https://create.kahoot.it/details/4b14f751-21e2-4baf-95ba-6b5e94ccd78e?drawer=